

Pharmacodynamics

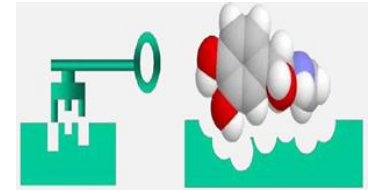
Pharmacodynamics

- Drug receptors & signaling mechanisms
- Agonists & antagonists
- Receptor regulation
- Factors affecting drug response & dosage

Pharmacodynamics (PD)

- The study of biochemical and physiological effects of drugs and the molecular mechanisms by which effects are produced
- What a drug does to the body and how it does (mechanisms of action - MoA)
- Effects produced by drugs could be by physical effect, chemical effect, or target molecule binding mediated effect (receptors, enzymes, ion channels, transporters)

Pharmacodynamics



- A receptor is component of a cell that interacts with a drug & initiates chain of biochemical events leading to observed effects
- Properties of receptors
 - regulatory macromolecules (mostly proteins, recognition sites)
 - most have active & inactive states; they are selective
 - up- or down regulated (supersensitivity, desensitization)

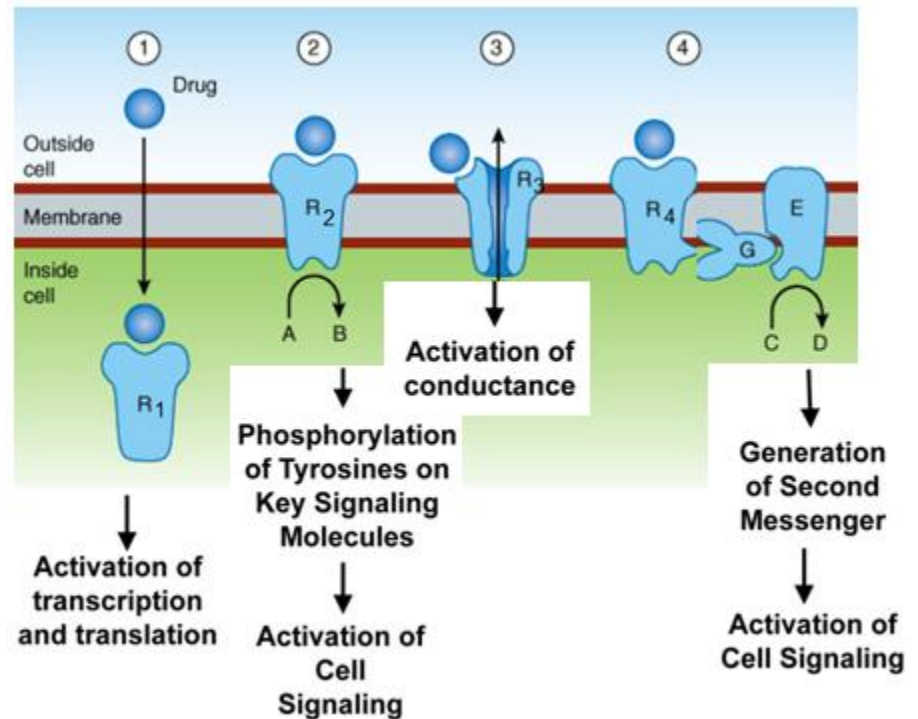
Receptor concepts

- ✓ The receptor's affinity for binding a drug determines the conc. of drug required to form D-R complexes
- ✓ Response to drugs is proportional to proportion of receptors occupied. Not all receptors need necessarily to be occupied by an active agonist in order to give a maximal response
- ✓ The free receptors remain (other than needed to produce max. response) are known as spare receptors

Pharmacodynamics...

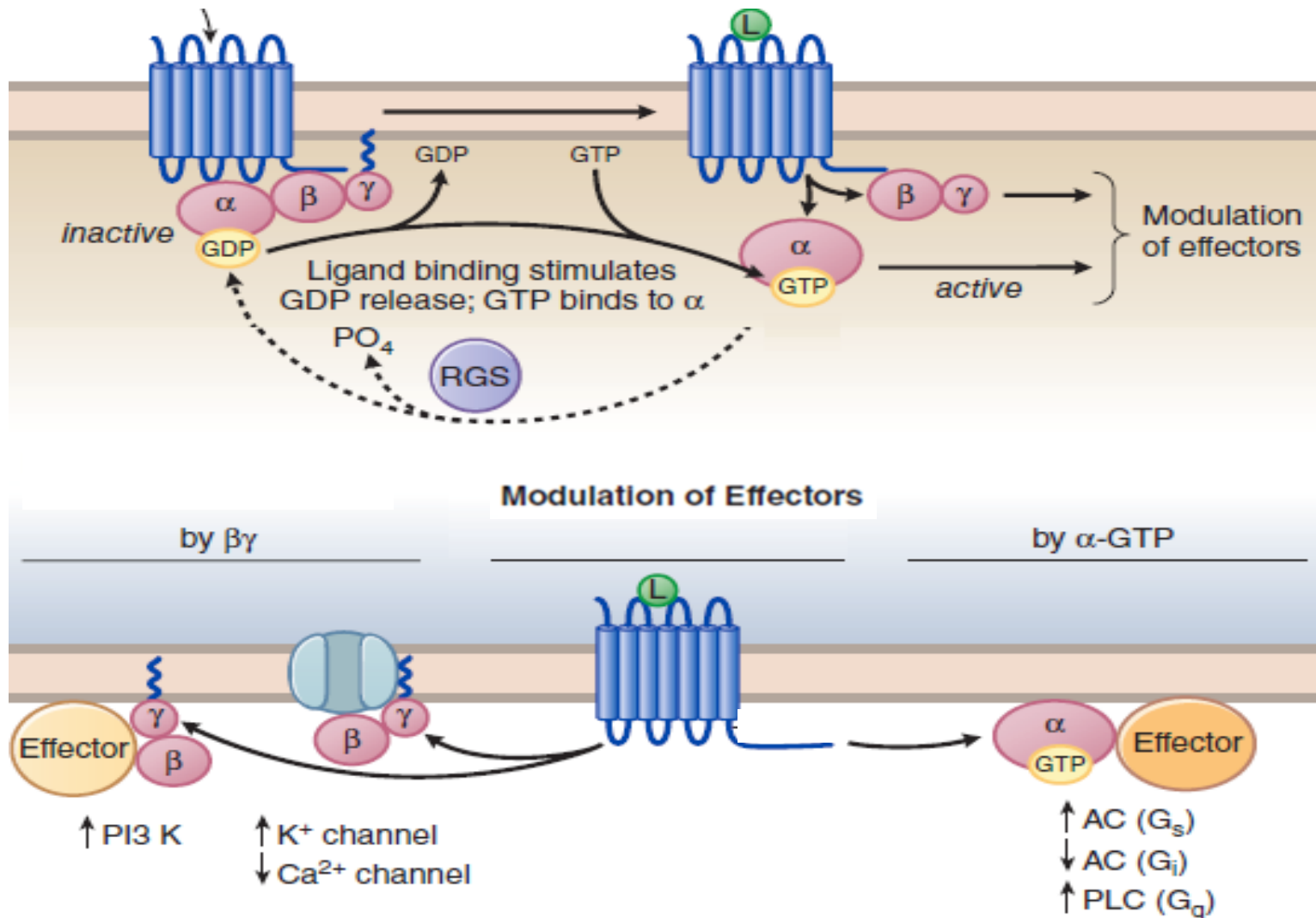
- Receptor subtypes for signal transduction

- Intracellular
- Enzyme linked
- Ion channel coupled
- G-protein coupled (GPCR)



G-Protein-Coupled Receptors...

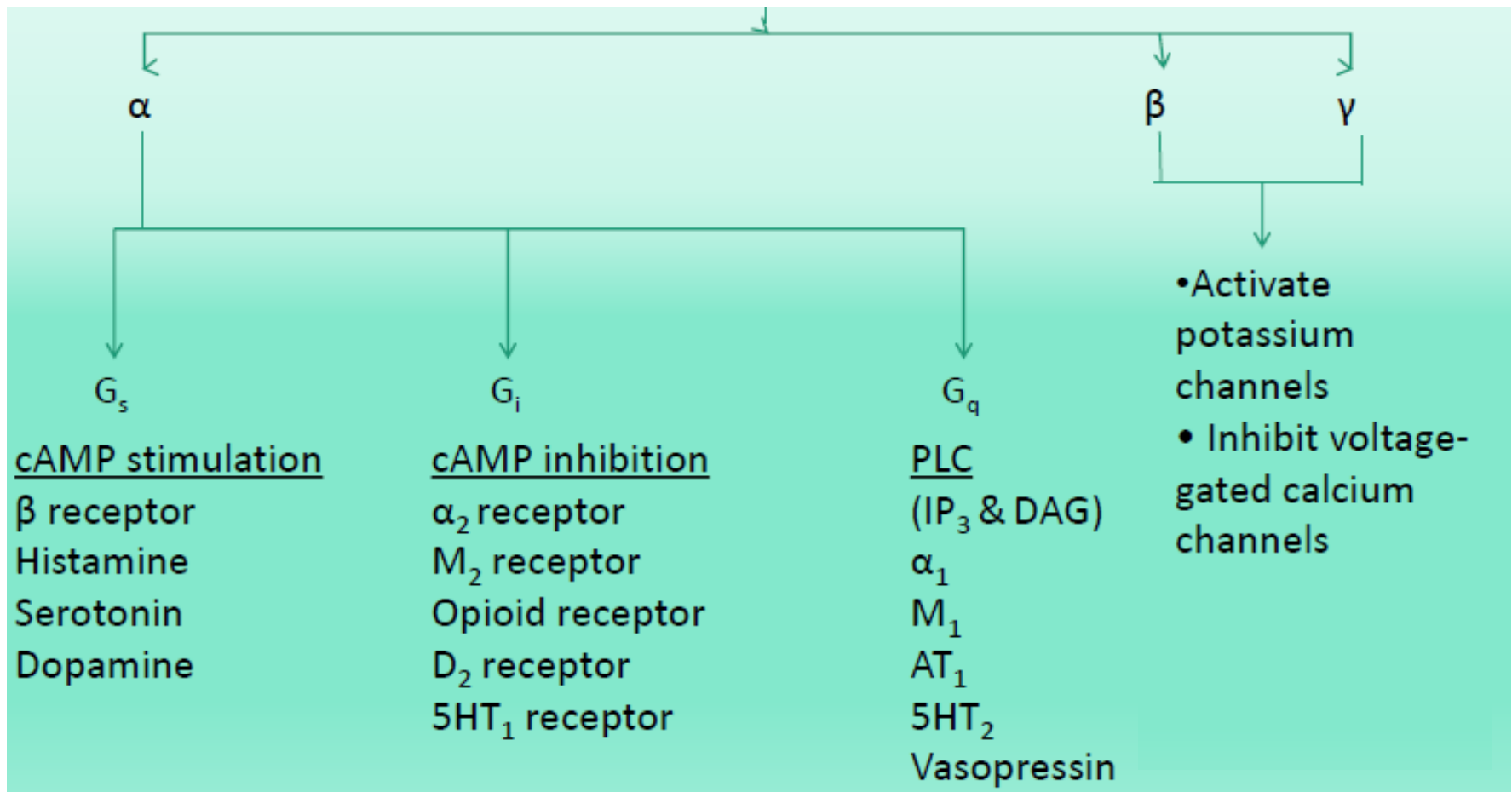
- GPCRs control different aspects of cell function (main targets)
 - **Adenylyl cyclase:** enzyme responsible for cAMP formation
 - **Phospholipase-C:** enzyme responsible for inositol triphosphate (IP3) and diacylglycerol (DAG) formation
 - **Ion channels:** particularly Ca^{2+} & K^{+} channels



GPCR receptor-effector pathway

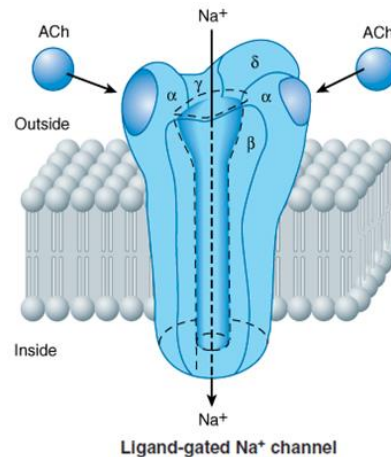
AC: adenylyl cyclase; PI3K: phosphatidylinositol 3-kinase; PLC: phospholipase-C;
RGS: regulator of G-protein signaling

G-Protein subunits with second messenger

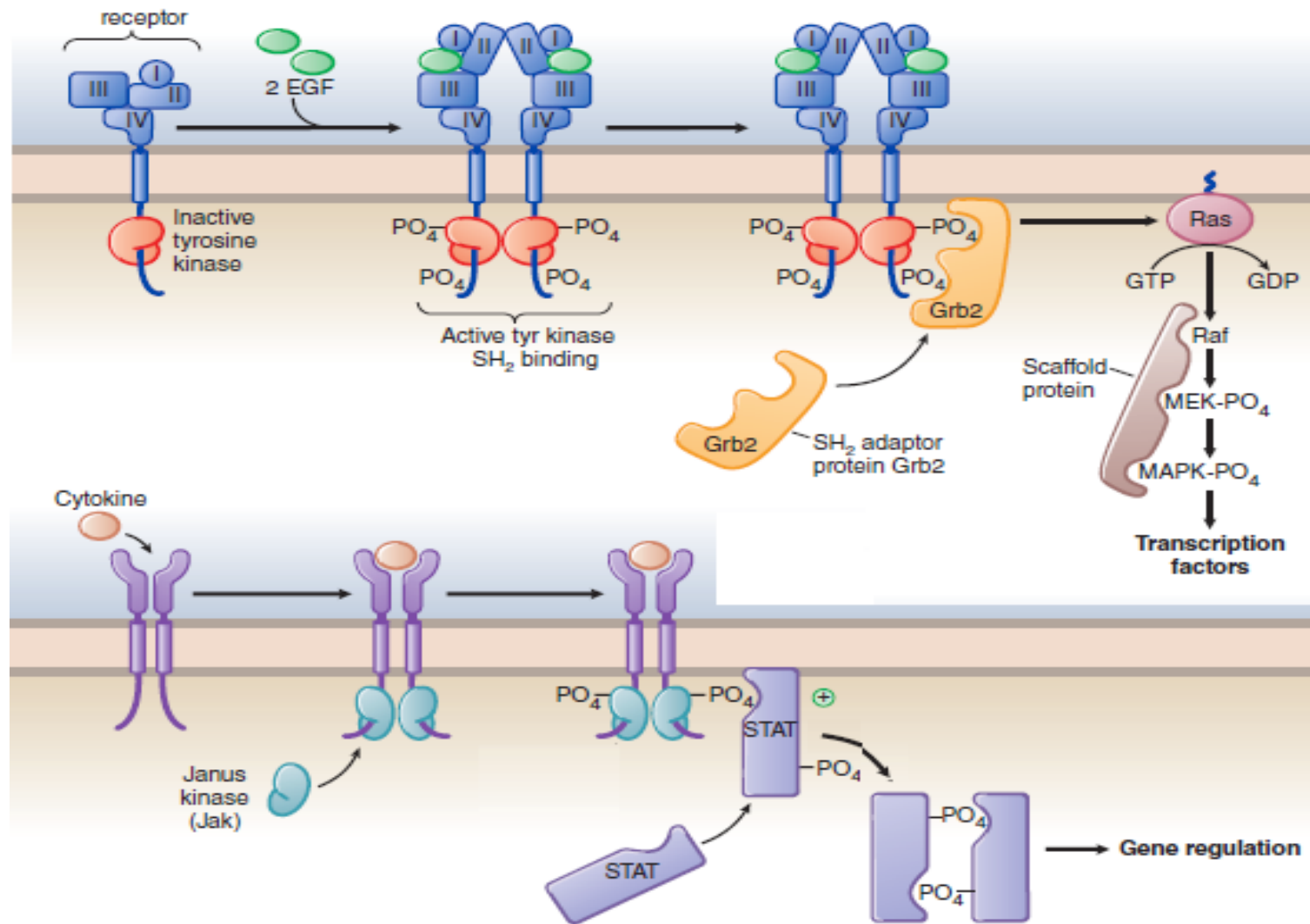


Ligand-gated ion channels

- Most involved in the nervous system
 - nicotinic acetylcholine (nAChR), GABA_A, glycine, glutamate
- Others: - IP₃-sensitive Ca²⁺ channel on ER
 - K_{ATP} channel in pancreatic β -cells

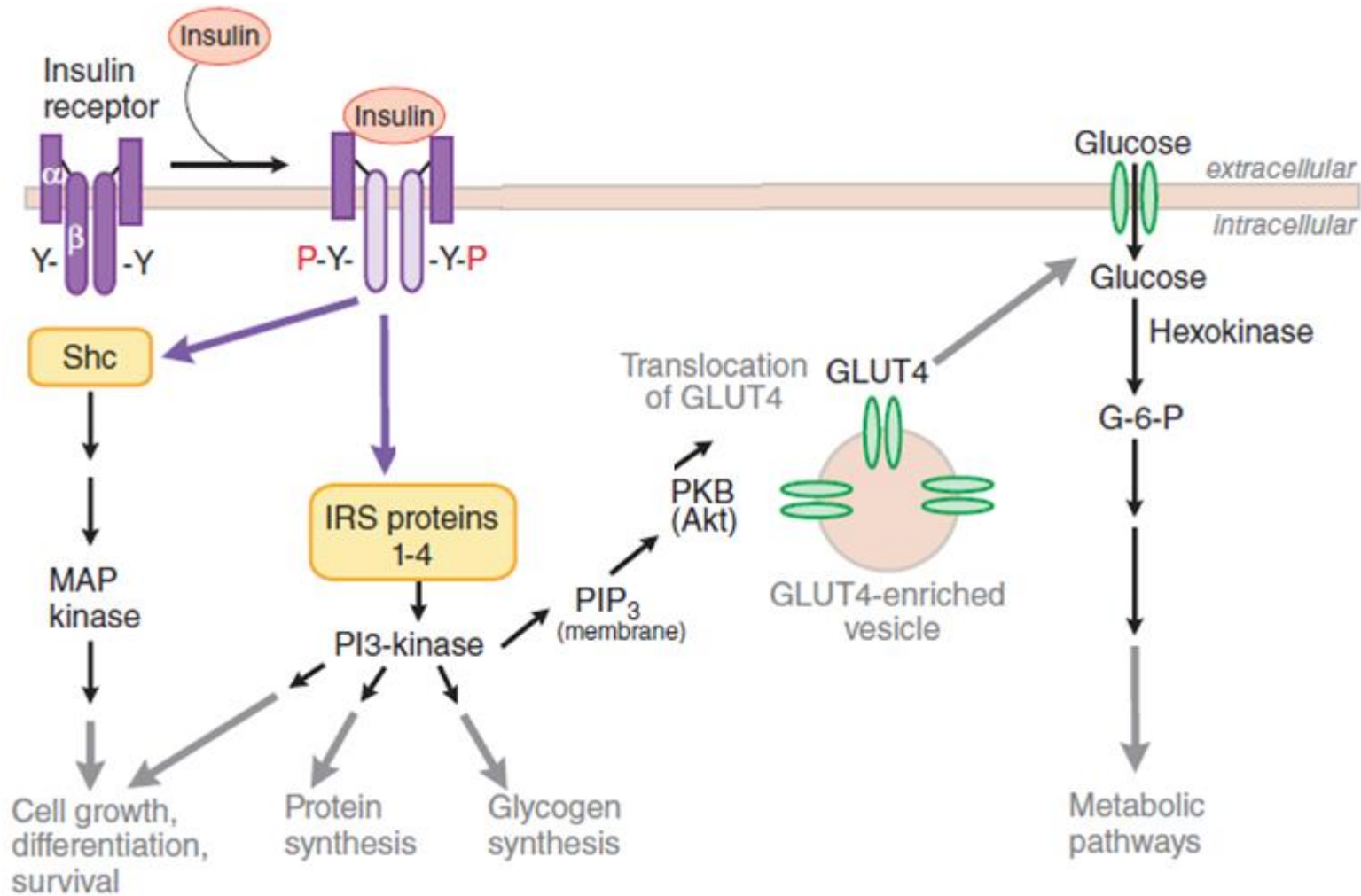


Kinase-linked receptors



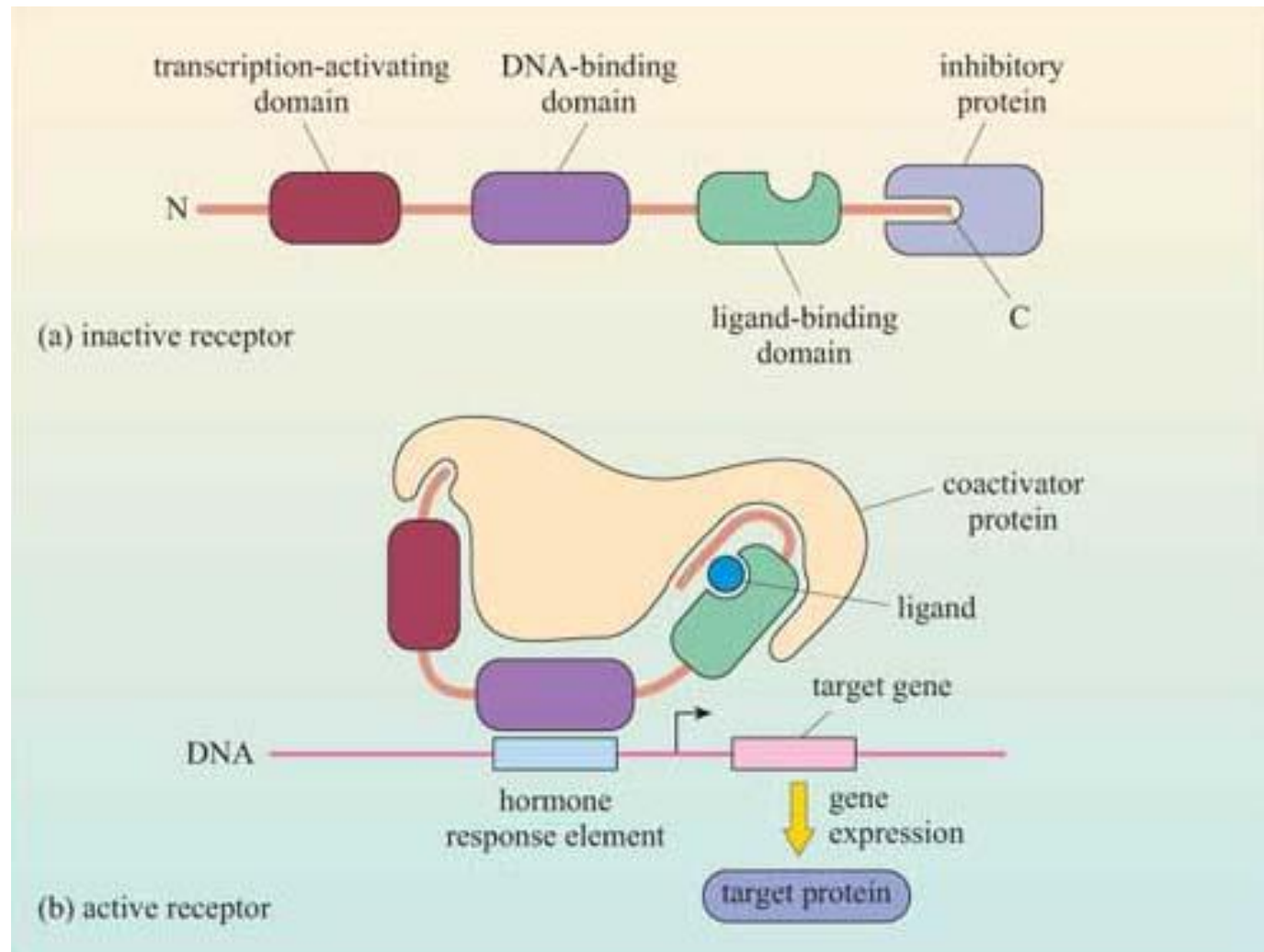
Activation of receptor tyrosine kinase: EGF: Epidermal growth factor (GF); Grb2: GF receptor-bound protein; MAPK: Mitogen-activated protein kinase; Ras: monomeric small G-protein; SH2: Src homology 2 domain; STAT: signal transducer and activator of transcription

Insulin receptor



Nuclear receptors (NR)

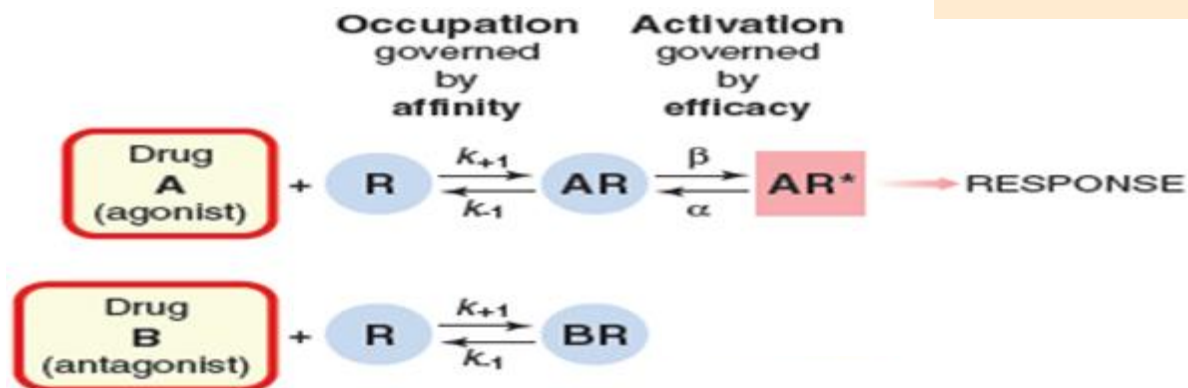
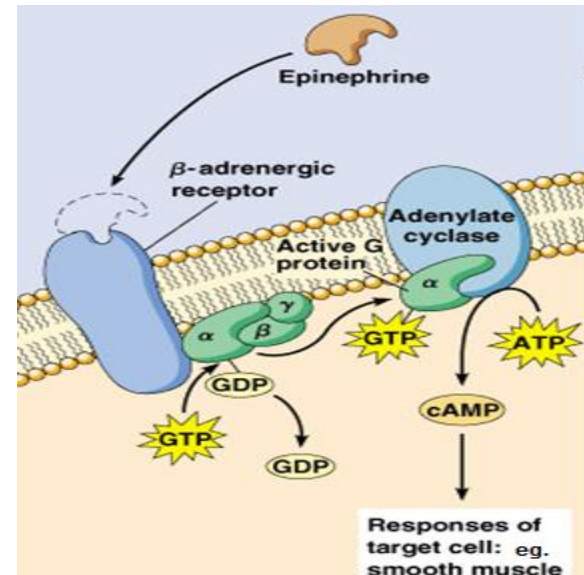
- Reside in cytoplasm/nucleus
- Ligands include steroids (glucocorticoids, mineralocorticoids, sex steroids, vitamin-D; thyroid hormone, retinoic acid,...
- Influence a large no. of genes: regulate gene expression, cellular differentiation, and control many physiol. processes



Activation of a nuclear hormone receptor

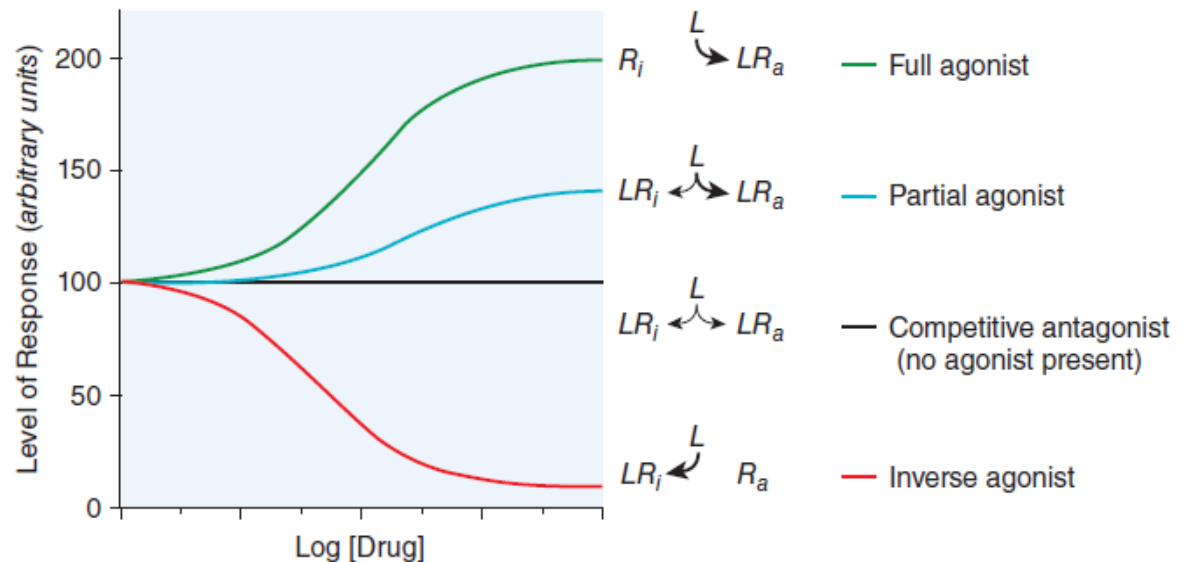
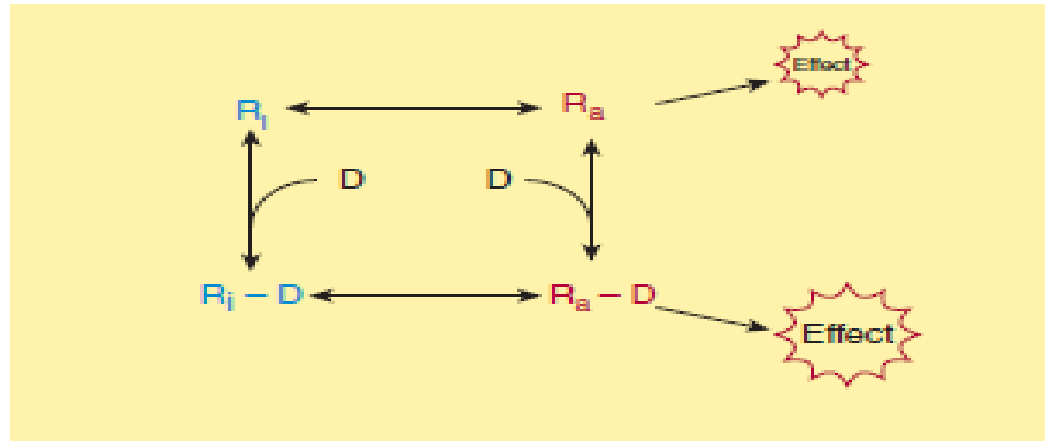
Pharmacodynamics...

- Based on receptor mediated effect drugs can be classified as
 - Agonists
 - Antagonists



Pharmacodynamics...

- Agonists could be
 - Full agonists
 - Partial agonists
 - Inverse agonists



Receptor activity regulation with conformation & drug selectivity

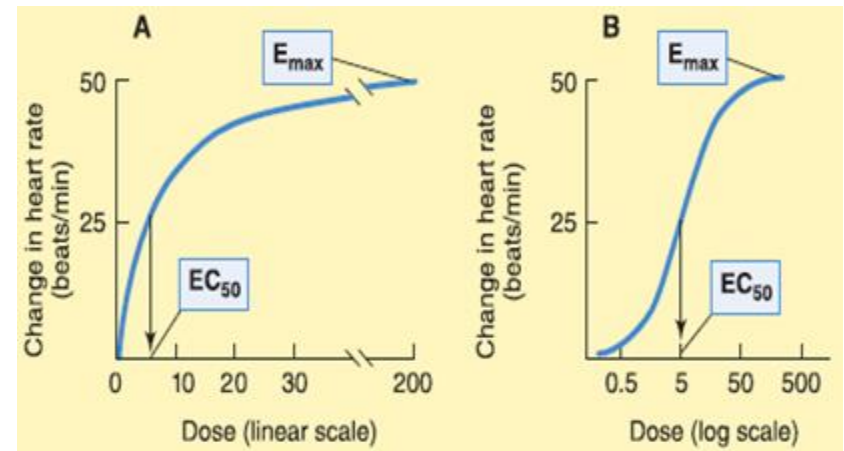
Dose-response relationship

- It is the relationship b/n intensity of drug effect and drug dose or concentration
- Concn-effect or dose-response curve is useful to estimate the
 - ✓ maximal response that the drug can produce ($E_{\max}$)
 - ✓ concn or dose needed to produce 50% of the maximal response (EC_{50} or ED_{50})

Types of DRCs

- **Graded DRC**

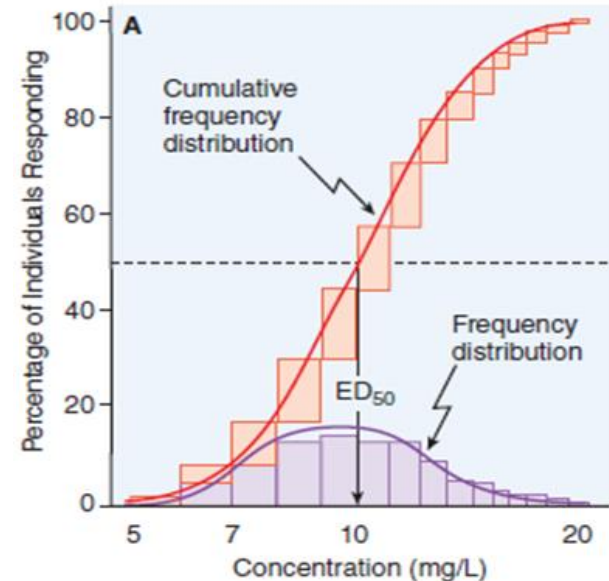
- graph of response to drug concn/dose
 - magnitude of response is graded
- used for assessment of conc. Vs effect in a person, in a tissue or in an experimental animal
- also useful to determine maximal efficacy of a drug



Types of DRCs...

- **Quantal DRC**

- graph of fraction/percent of the population responders with doses
- shows cumulative frequency of occurrence of specific response
- response may be therapeutic or toxic effect
- useful to determine therapeutic index and compare safety

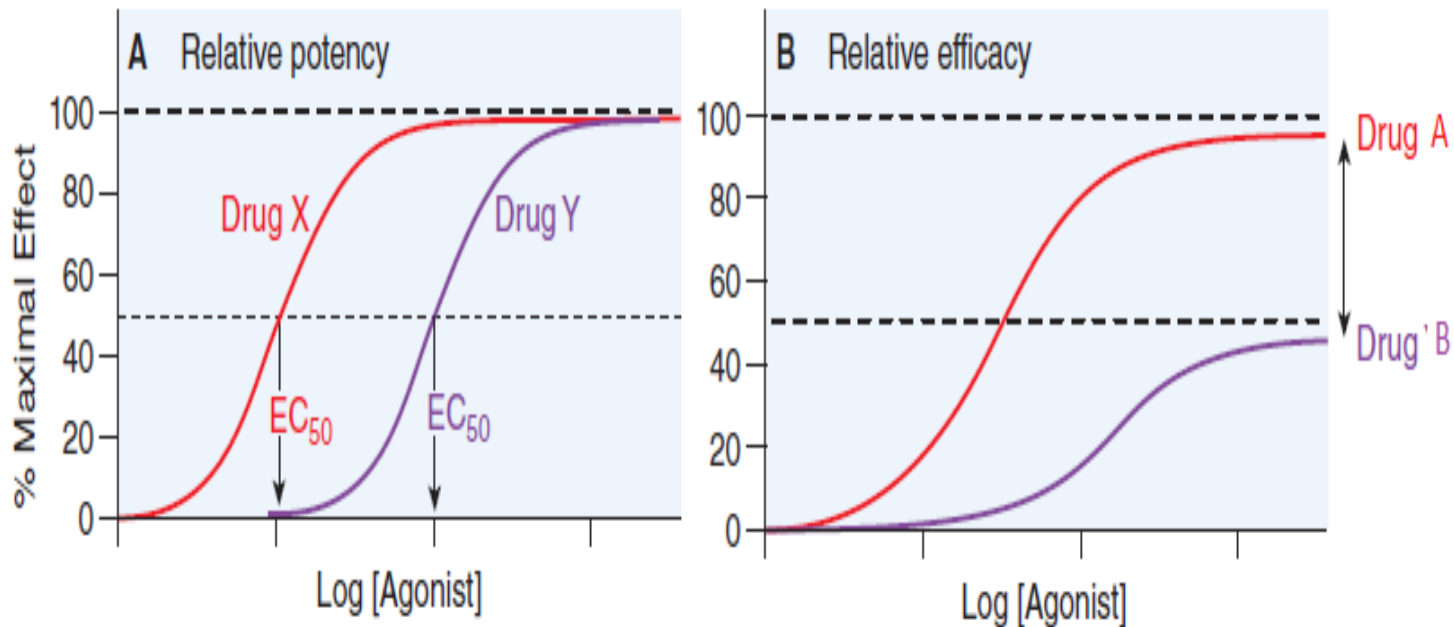
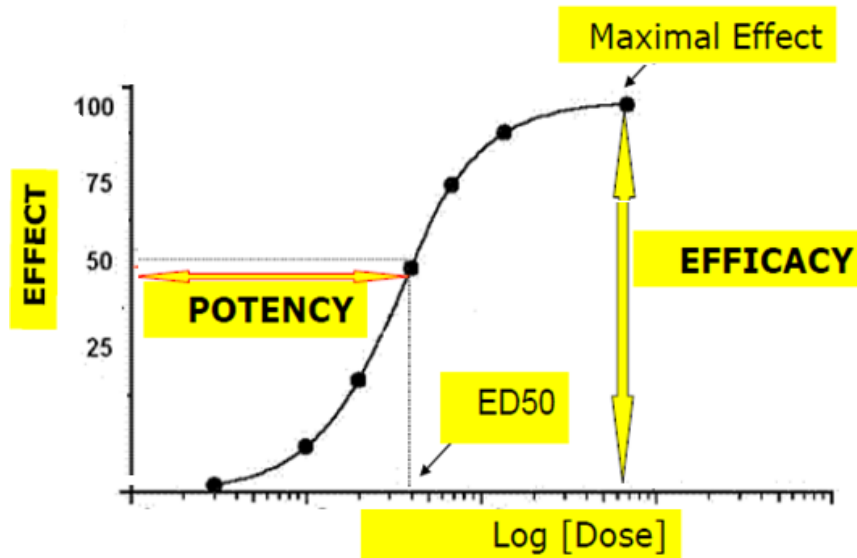


Intrinsic Activity (IA)

- IA is the ability of a drug to initiate a response
- A mathematical value that indicates efficiency with which drug-receptor interaction is coupled to response
- Determines the efficacy of a drug
 - Full agonist ($IA = 1$, Produce maximal effects)
 - Partial agonist ($0 < IA < 1$, produce sub-maximal effects)
 - Antagonist ($IA = 0$, no efficacy)

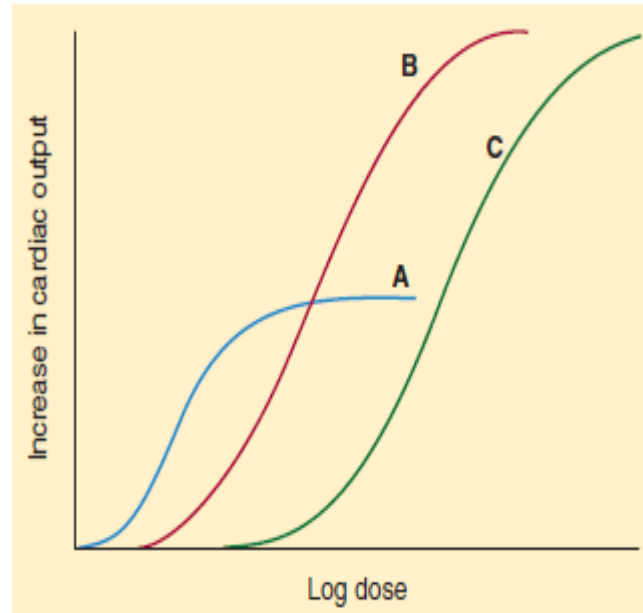
Potency and Efficacy

- **Efficacy:** ability of a drug to produce the maximum effect
 - A measure of the efficiency of coupling the drug–receptor complex with a pharmacological response
 - relative efficacy of two drugs that elicit the same effect can be measured by comparing the maximum effects of the drugs
- **Potency:** related to size of the dose necessary to produce effect
 - influenced by its affinity for its receptor



ED_{50} - effective dose that produces 50% of the maximum response

A study was carried out to compare two new drugs (A & B) to a standard agent (C). The results of cardiac output measurements are shown in the graph below.



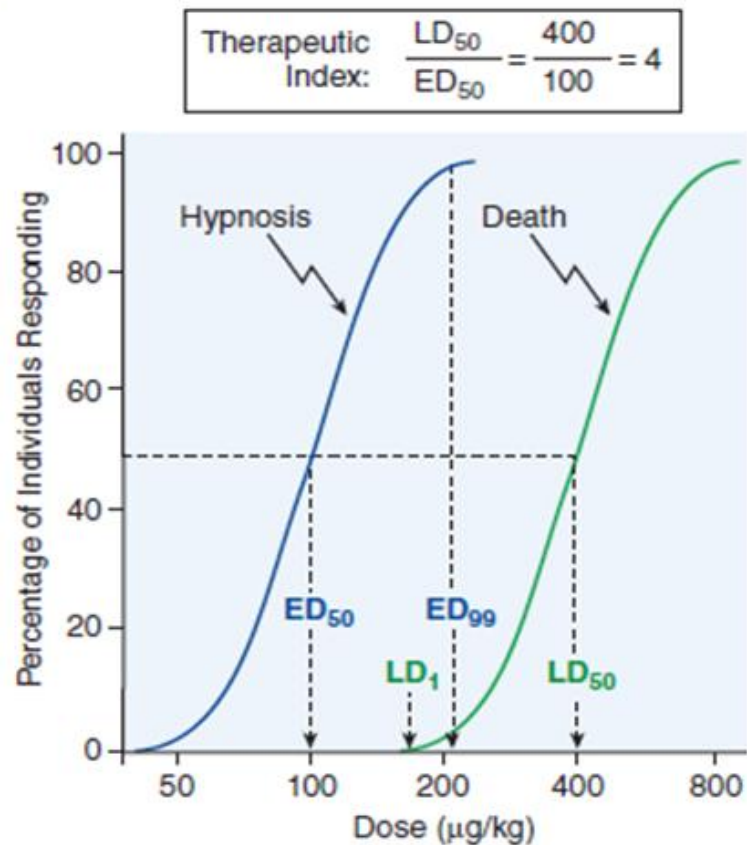
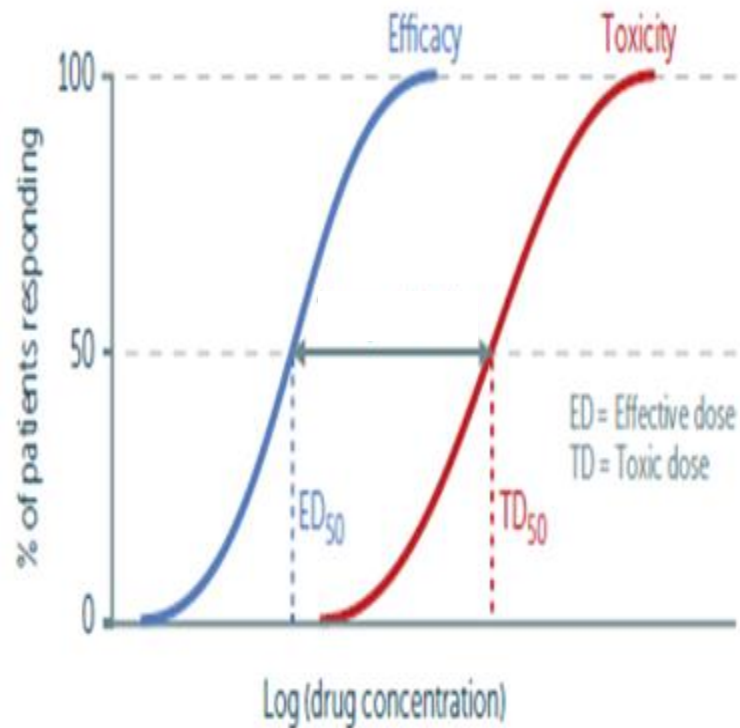
Which one of the following statements is correct?

- a. Drug A is the least potent but the most effective drug
- b. Drug B is more potent but less effective than drug C
- c. Drug C is less effective but more potent than drug B
- d. Drug B is more effective than drug A

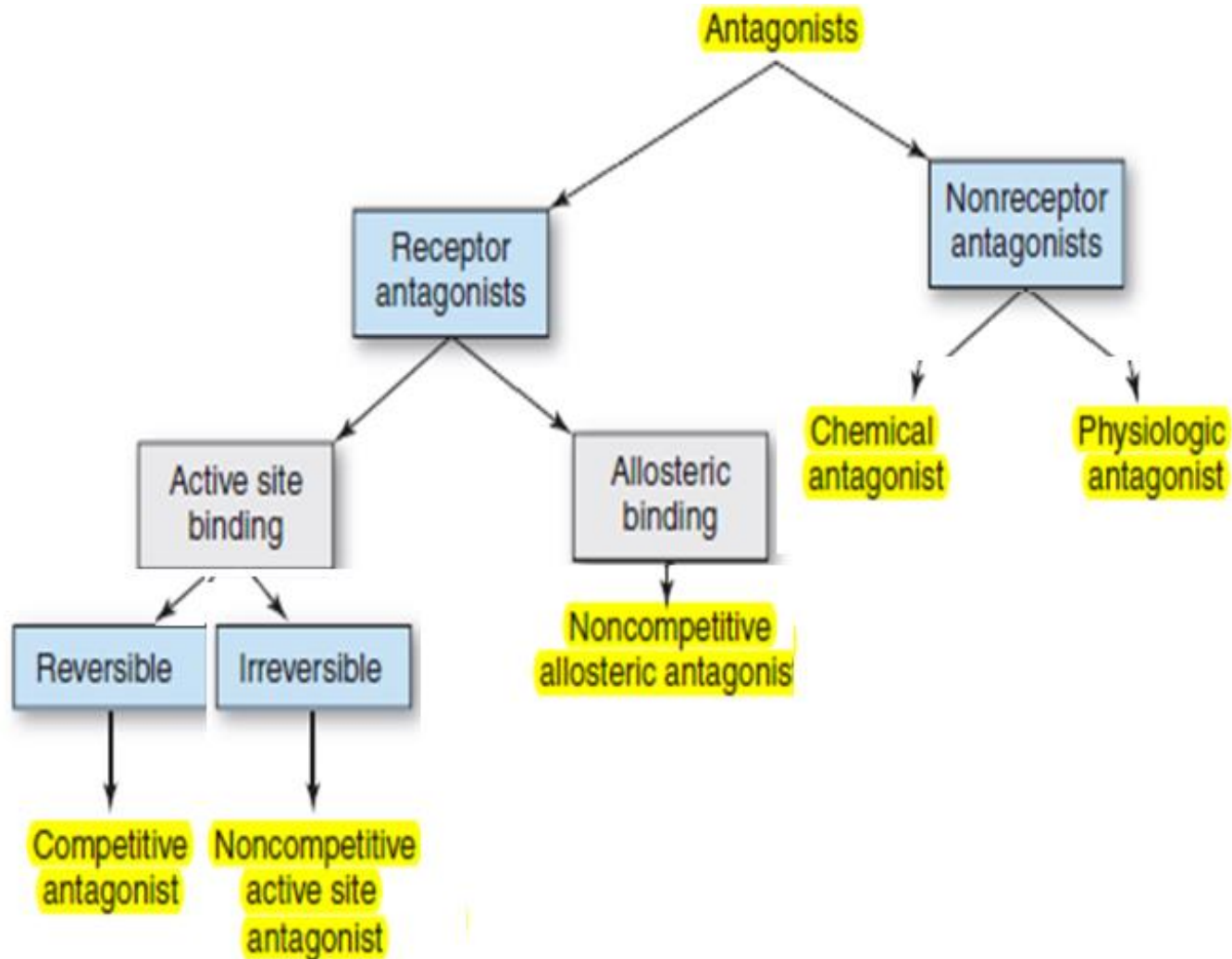
Therapeutic index

Measurement of drug safety.

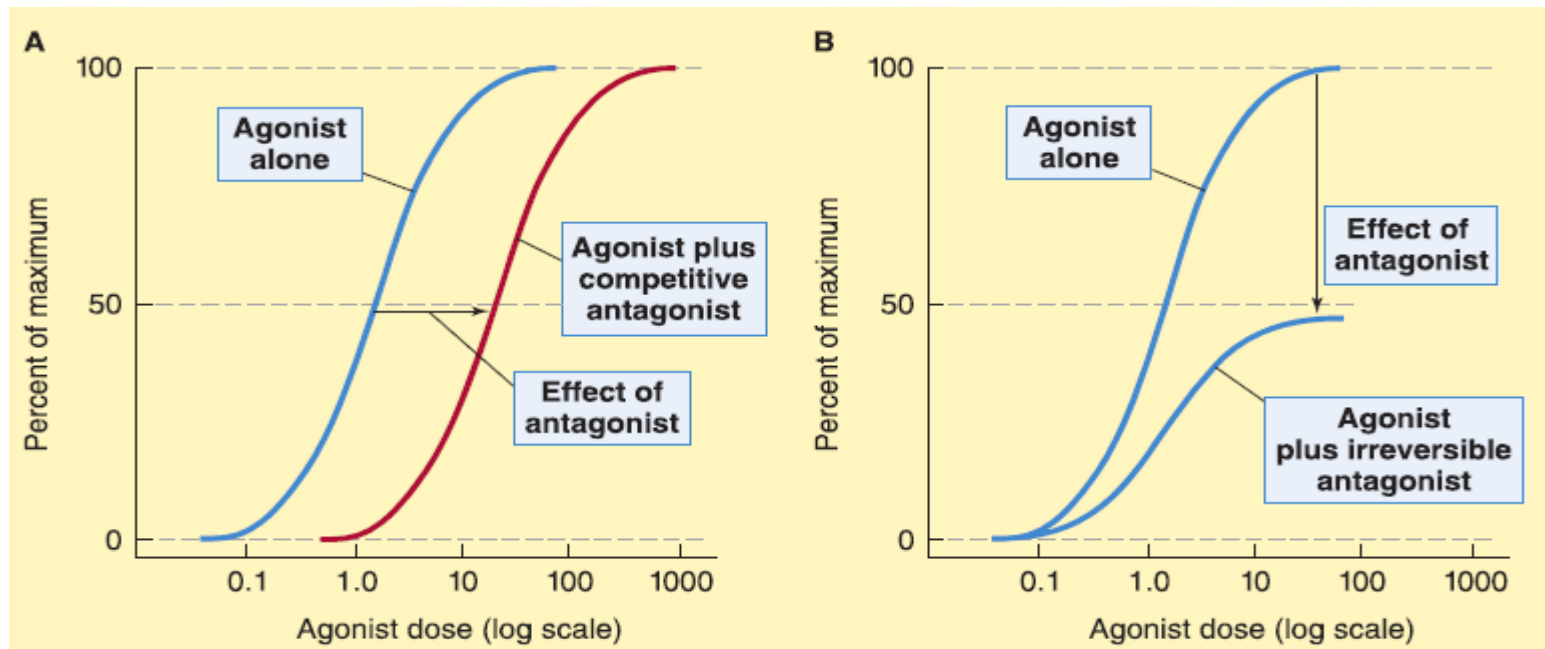
$$\frac{TD_{50}}{ED_{50}} = \frac{\text{median toxic dose}}{\text{median effective dose}}$$



Pharmacodynamics...



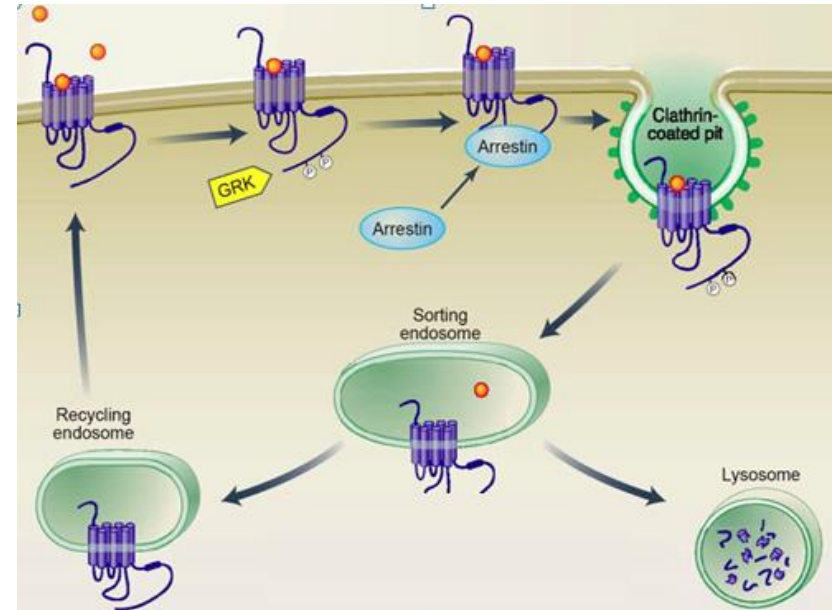
Pharmacodynamics...



AGONIST WITH	EFFICACY	REMARKS	EXAMPLE
A Competitive antagonist	No change	Can be overcome by ↑ agonist concentration	Diazepam (agonist) + flumazenil (competitive antagonist) on GABA _A receptor.
B Noncompetitive antagonist	↓	Cannot be overcome by ↑ agonist concentration	Norepinephrine (agonist) + phenoxybenzamine (noncompetitive antagonist) on α-receptors.

Receptor regulation

- Sensitization or up-regulation
 - ✓ Continuous/prolonged use of receptor blocker drug
 - ✓ Inhibition of syn./ release of hormone/ transmitters



- Desensitization or down-regulation
 - ✓ Prolonged/continuous use of agonist
 - ✓ Inhibition of degradation or uptake of agonist

Factors that modify drug effect & drugs dosage

- Individual factors

- Age

$$\text{Young's Rule} = \text{Child dose} = \frac{\text{Age in year} \times \text{adult dose}}{\text{Age} + 12}$$

- Sex

$$\text{Clark's rule} = \text{Child dose} = \frac{\text{Weight in lb} \times \text{adult dose}}{150}$$

- body mass index

$$\text{Child dose} = \frac{\text{Body surface area of child} \times \text{adult dose}}{\text{Body surface area of adult}}$$

- Habits

$$\text{Surface area in m}^2 = \frac{\sqrt{\text{height} \times \text{weight}}}{60}$$

- genetics

Factors that modify drug effect & drugs dosage

- Route and time of drug admn
- Presence of chronic diseases
- Drug interactions (D/I): Pk & Pd levels
 - Types: addition, potentiation, synergism, antagonism

Summary

- Pharmacodynamics (pd) refers to effects of a drug on the body
- Most drugs act by binding to physiological receptors and alter the rate or magnitude of an intrinsic cellular response
- Important pd properties of a drug: affinity, efficacy & specificity
- effects of a drug depend on many pd factors, including receptor specificity, tissue-specific expression of the receptor(s), drug concn in receptor compartment on target tissues, genetics, and interactions with other drugs at the receptor